Question 1:
How many stars will the following snippet print to the monitor?

```
i = 0
while i <= 5:
    i += 1
    if i % 2 == 0:
        break
    print('*')
```

A. zero
B. two
C. one
D. three

Question 2:
Which of the following statements is false?

A. The None value can not be used as an argument of arithmetic operators.
B. The None value may not be used outside functions.
C. The None value can be compared with variables.
D. The None value can be assigned to variables.

Question 3:
What is the expected output of the following code?

```
print(1 / 1)
```

A. 1.0
B. 1
C. This can not be predicted.
D. This can not be evaluated.

Question 5:
The += operator, when applied to strings, performs:

A. Multiplication
B. Concatenation
C. Subtraction
D. None of the above.

Question 6:
What is the expected output of the following code?

```
data = [
    [1, 2, 3, 4],
    [5, 6, 7, 8],
    [9, 10, 11, 12],
    [13, 14, 15, 16]
]
for i in range(0, 4):
    print(data[i].pop(), end=' ')
```

A. 1 5 9 13
B. 13 14 15 16
C. 1 2 3 4
D. 4 8 12 16

Question 7:
What is the expected output of the following code?

```
data = {'1': '0', '0': '1'}

for d in data.vals():
    print(d, end=' ')
```

A. 0 0
B. 1 0
C. 0 1
D. The code is erroneous.

Question 8:
If you want to import pi from math, which line will you use?

A. from pi import maths
B. import pi from maths
C. from maths import pi
D. None of the above.

Question 9:
What is the expected output of the following code?

```
try:
    print('try')
except:
    print('except')
finally:
    print('finally')
```

A.	The code is erroneous.
B.	1 | finally 2 | try
C.	1 | finally 2 | except
D.	1 | try 2 | finally
E.	1 | except 2 | finally

Question 10:

Which of the following sentences correctly describes the output of the below Python code?

```
data = [4, 2, 3, 2, 1]
res = data[0]

for d in data:
    if d < res:
        res = d

print(res)
```

A. res is the sum of all the numbers in the list.
B. res is the smallest number in the list.
C. res is the largest number in the list.
D. t'es is the average of all the numbers in the list.
E. None of the above.

Question 11:

What is the expected output of the following code?

```
print(chr(ord('z') - 2))
```

A. a
B. x
C. z
D. y

Question 12:
What is the expected output of the following code?

```
with open('data.txt', 'w') as f:
    f.write("I'm gonna make him an offer he can't refuse.")

with open('data.txt', 'r') as f:
    data = f.readlines()
    for line in data:
        words = line.split()
        print(words)
```

A. I ‘m gonna make him an offer he can’t refuse.
B. [“I’m” , ‘gonna’ , ‘make’ , ‘him’ , ‘an’ , ‘offer’ , ‘he’ , “can;t” , ‘refuse’ .]
C. I
D. The code is erroneous.

Question 13:
Consider the following code.

```
x = 1
x = x == x
```

The value eventually assigned to x is equal to:

A. 0
B. False
C. 1
D. True

Question 14:
Consider the following code.

```
from x.y import z
```

The code causes the import of ...

A. entity z from module y from package x
B. entity z from module from package y
C. entity x from module y from package z
D. entity y from module x from package z

Question 15:
What is the expected output of the following code?

```
data = 'abbabadaadbbaccabc'
print(data.count('ab', 1))
```

A. 2
B. 3
C. 4
D. 5

Question 16:
Which of the following statements is false?

A. The result of the / operator is always an integer value.
B. The right argument of the % operator can not be zero.
C. Multiplication precedes addition.
D. The ** operator has right-to-left associativity.

Question 17:
What is the expected output of the following code?

```
a = [1, 2, 3, 4, 5]
print(a[3:0:-1])
```

A. [4, 3, 2]
B. [4, 3]
C. The code is erroneous.
D. [4, 3, 2, 1]

Question 18:
What is the expected output of the following code?

```
people = {}

def add_person(index):
    if index in people:
        people[index] += 1
    else:
        people[index] = 1

add_person('Peter')
add_person('Paul')
add_person('peter')

print(len(people))
```

A. 3
B. The code is erroneous.
C. 2
D. 1

Question 19:
You develop a Python application for your company. A list named employees contains 200 employee names, the last five being company management. You need to slice the list to display all employees excluding management. Which code segments can you use? Choose two.

A. employees [: -5]
B. employees [0: -5]
C. employees [1: -5]
D. employees [0: -4]
E. employees [1: -4]

Question 20:
What is the expected output of the following code?

```
x = 9
y = 12
result = x // 2 * 2 / 2 + y % 2 ** 3
print(result)
```

A. 7.0
B. 8.0
C. 9.0
D. 8

Question 21:
What is the expected output of the following code?

```
data = [[0, 1, 2, 3] for i in range(2)]
print(data[2][0])
```

A. 2
B. 1
C. 0
D. The code is erroneous.

Question 22:
What is the expected output of the following code?

```
data = set([1, 2, 2, 3, 3, 3, 4, 4, 4, 4])
print(len(data))
```

A. 0
B. 1
C. 4
D. 2
E. 3
F. 10

Question 23:
Consider the following code.

```
import random
data = [10, 20, 30]
random.shuffle(data)
print(data)
```

Which of the following statements best describes the behaviour of the random.shuffle() method?

A. It shuffles the elements for the number of times equal to the size of the list.
B. It will not modify the list. This function is just a placeholder and yet to be implemented.
C. It shuffles the elements of the list in-place.
D. It returns a list where the elements 10, 20, and 30,would be at random positions.

Question 24:
A code point is:

A. A point used to write a code.
B. Acode containing a point.
C. A number which makes up a character.
D. None of the above.

Question 25:
The following statement ...

```
assert x == 0
```

A. will stop the program if x is not equal to 0
B. has no effect.
C. will stop the program if x is equal to 0
D. is erroneous.

Question 26:
What is the expected output of the following code?

```
print(1 // 2 * 3)
```

A. 4.5
B. 0
C. 0.0
D. 0.16666666666666666

Question 27:

The part of your code where the handling of an exception takes place should be placed inside:

A. the try: branch
B. the except: branch
C. the exception: branch
D. None of the above.

Question 28:
Consider the following code.

```
for n in range(1, 6, 1):
    print(??? * 5)
```

What would you insert instead ??? so that the program prints the following pattern to the monitor?

```
11111
22222
33333
44444
55555
```

A. n
B. str(n)
C. -1
D. 1
E. 2

Question 29:
How many stars will the following code print to the monitor?

```
x = 1
while x < 10:
    print('*')
    x = x << 1
```

A. one
B. eight
C. four
D. two

Question 30:
UTF-8 is ...

A. an encoding form of the Unicode Standard.
B. the 9th version of the UTF Standard.
C. a Python version name.
D. a synonym for “byte’

Question 31:
Which of the following function headers is correct?

A. def func (a=1, b=1, c=2):
B. def func (a =1, b):
C. def func (a=1, b=1, c=2, d):
D. def func (a=1,b, c=2):

Question 32:
What is the expected output of the following code?

```
colors = ['red\n', 'yellow\n', 'blue\n']
file = open('wellert.txt', 'w+')
file.writelines(colors)
file.close()
file.seek(0)
for line in file:
    print(line)
```

A.	redyellowblue
B.	The code is erroneous.
C.	[‘red\n’ , ‘yellow\n’ , ‘blue\n’]
D.	1 | red 2 | 3 | yellow 4 | 5 | blue

Question 33:
What is the expected output of the following code?

```
import os

os.mkdir('thumbnails')
os.chdir('thumbnails')

sizes = ['small', 'medium', 'large']

for size in sizes:
    os.mkdir(size)

print(os.listdir())
```

A. [‘large’ , ‘medium’ , ‘small’]
B. [ ]
C. [‘.’ , ‘. . ‘ ‘large’ , ‘medium’ , ‘small’]
D. [‘.’ , ‘large’ , ‘medium’ , ‘small’]

Question 34:
Which of the following items are present in the function header?

A. function name and parameter list
B. return value
C. function name
D. parameter list

Question 35:
What is the expected output of the following code?

```
try:
    raise Exception(1, 2, 3)
except Exception as e:
    print(len(e.args))
```

A. 1
B. The code is erroneous.
C. 2
D. 3

Question 37:
What is the expected output of the following code?

```
data = [1, 2, 3, 4, 5, 6]

for i in range(1, 6):
    data[i - 1] = data[i]

for i in range(0, 6):
    print(data[i], end=' ')
```

A. 1 2 3 4 5 6
B. 2 3 4 5 6 6
C. 1 1 2 3 4 5
D. 2 3 4 5 6 1

Question 38:
What is the expected output of the following code if the user enters 11 and 4 ?

```
x = int(input())
y = int(input())
x = x % y
x = x % y
y = y % x
print(y)
```

A. 2
B. 3
C. 4
D. 1

Question 39:
What is the expected result of the following code?

```
b = bytearray(3)
print(b)
```

A. 3
B. bytearray (b '3')
C. bytearray (0, 0, 0)
D. bytearray (b' \x00\x00\x00')

Question 40:
What is the expected output of the following code?

```
x = '\\\\'
print(len(x))
```

A. 1
B. The code is erroneous.
C. 2
D. 4

Question 41:
What is the expected behaviour of the following snippet?

```
1  x = 1
2
3
4  def a(x):
5      return 2 * x
6
7
8  x = 2 + a(x)        # Line 8
9  print(a(x))         # Line 9
```

It will:

A. print 4
B. print 6
C. cause a runtime exception on Line 8
D. print 8
E. cause a runtime exception on Line 9

Question 42:
What is the expected output of the following code?

```
data = 'abcdefg'

def func(text):
    del text[2]
    return text

print(func(data))
```

A. abcef
B. acdef
C. abdef
D. The code is erroneous.

Question 43:
Which of the following function calls can be used to invoke the below function definition?

```
def test(a, b, c, d):
```

Choose three.

A. test (1, 2, 3, d=4)
B. test (a=1, b=2, c=3, d=4)
C. test (a=1, 2, 3, 4)
D. test (a=1, b=2, c=3, 4)
E. test (1, 2, 3, 4)
F. test (a=1,2, c=3, 4)

Question 44:
How many stars will the following code send to the monitor?

```
x = 0
while x < 6:
    x += 1
    if x % 2 == 0:
        continue
    print('*')
```

A. two
B. three
C. one
D. zero

Question 45:
What is the expected output of the following code?

```
def func(message, num=1):
    print(message * num)

func('Hello')
func('Welcome', 3)
```

A.	1 | Hello 2 | Welcome Welcome Welcome
B.	1 | Hello 2 | WelcomeWelcomeWelcome
C.	1 | Hello 2 | Viewers
D.	1 | Hello 2 | Welcome, Welcome, Welcome

Question 46:
What is the expected output of the following code?

```
x = True
y = False
x = x or y
y = x and y
x = x or y
print(x, y)
```

A. False False
B. False True
C. True True
D. True False

Question 47:
What is the expected output of the following code?

```
print('Peter' 'Wellert')
```

A. The code is erroneous.
B. Wellert
C. PeterWellert
D. Peter

Question 48:
Which of the following variable names is illegal?

A. tRUE
B. TRUE
C. True
D. true

Question 49:
What is the expected output of the following code?

```
print(float('1.3'))
```

A. 13
B. 1.3
C. The code is erroneous.
D. 1, 3

Question 50:
How many stars will the following code print to the monitor?

```
i = 0
while i < i + 2:
    i += 1
    print('*')
else:
    print('*')
```

A. The snippet will enter an infinite loop.
B. two
C. zero
D. one

Question 51:
What is the expected output of the following code?

```
x = 1 + 1 // 2 + 1 / 2 + 2
```

```
print(x)
```

A. 4.0
B. 4
C. 3
D. 3.5

Question 52:
What is the expected output of the following code?

```
data = {}

def func(d, key, value):
    d[key] = value

print(func(data, '1', 'Peter'))
```

A. None
B. value
C. Peter
D. 1
E. The code is erroneous.

Question 53:

The ABC Video company needs a way to determine the cost that a customer will pay for renting a DVD. The cost is dependent on the time of day the DVD is returned. However, there are also special rates on Thursdays and Sundays. The fee structure is shown in the following list:

The cost is $1.59 per night. If the DVD is returned after 8 PM, the customer will be charged an extra day. If the video is rented on a Sunday, the customer gets 30% off for as long as they keep the video. If the video is rented on a Thursday, the customer gets 50% off for as long as they keep the video.

You need to write code to meet the requirements.

```
# ABC Video, DVD Rental Calculator

ontime = input('Was the video returned before 8 pm? y or n').lower()
days_rented = int(input('How many days was the video rented?'))
day_rented = input('What day was the video rented?').capitalize()

cost_per_day = 1.59

if ontime XXX
    days_rented += 1
if day_rented YYY
    total = (days_rented * cost_per_day) * .7
elif day_rented ZZZ
    total = (days_rented * cost_per_day) * .5
else:
    total = (days_rented * cost_per_day)

print('Cost of the DVD rental is: $', total)
```

What should you insert instead of XXX, YYY and ZZZ?

A.	1 | XXX -> == 'y' : 2 | YYY -> == 'Sunday' : 3 | ZZZ -> == 'Thursday' :
B.	1 | XXX -> != 'n' : 2 | YYY -> is 'Sunday' : 3 | ZZZ -> is 'Thursday' :
C.	1 | XXX -> == 'n': 2 | YYY -> is 'Sunday' : 3 | ZZZ -> is 'Thursday' :
D.	1 | XXX -> == 'n' : 2 | YYY -> 'Sunday' : 3 | ZZZ -> 'Thursday' :
E.	1 | XXX -> 'y' : 2 | YYY -> 'Sunday': 3 | ZZZ -> 'Thursday' :

Question 54:
Which of the following functions immediately terminates a program?

A. None
B. sys.halt ( )
C. sys.exit ( )
D. sys.stop ( )
E. sys.terminate ( )

Question 55:
Which of the following is false?

A. A try statement can have one or more final clauses.
B. A try statement can have a finally clause and an except clause.
C. A try statement can have one or more except clauses.
D. A try statement can have a final clause without an except clause.

Question 56:
What value will be assigned to the x variable?

```
z = 3
y = 7
x = y == z and y > z or z > y and z != y
```

A. True
B. 0
C. 1
D. False

Question 57:
What is the expected output of the following code?

```
num = 1

def func():
    num = 3
    print(num, end=' ')

func()

print(num)
```

A. 1 1
B. 1 3
C. 3 3
D. The code is erroneous.
E. 3 1